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ENGINEERING



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Student Performance Analysis Using Exploratory Data Analysis

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA 0404 - FUNDAMENTALS OF DATA SCIENCE
to the award of the degree of

BACHELOR OF ENGINEERING

IN

**COMPUTER SCIENCE ENGINEERING
(AI)**

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Student Performance Analysis Using Exploratory Data Analysis” has been carried out by Y Sravan Kumar (192472289) under the supervision of Dr. Krishna Raj Mony and is submitted in partial fulfilment of the requirements for the current semester of the B.E Computer Science Engineering (AI) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, [Y Sravan Kumar] [G. Mohammad Taher] [S. Siva Charan] of the [CSE AI] SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Student Performance Analysis Using Exploratory Data Analysis” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: SIMATS, Chennai – 602105.

Date: 01-09-2026

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Sravan Kumar Y / 192472289	Data collection, data cleaning and preprocessing	Designed and developed the data collection and preprocessing module	Tested the cleaned dataset and checked missing values and data consistency	Contributed to Module 1 and related documentation	33%
G. Mohammad Taher / 192524366	Performance prediction using machine learning	Developed the ML model for predicting student performance	Tested the model and analyzed prediction results and performance metrics	Contributed to Module 2 and related documentation	34%
S. Siva Charan / 192472147	Visualization and insight dashboard	Designed and developed charts, visualizations and dashboard	Tested dashboard outputs and analyzed student performance insights	Contributed to Module 3 and related documentation	33%
Total					100%

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ABSTRACT

Student performance is influenced by several factors such as marks, attendance, assignment scores, internal assessments, and previous academic results. Manually analyzing these factors can take considerable time and may make it difficult for teachers to identify students who are performing well or may require additional support. Therefore, there is a need for a simple data-driven system that can analyze student records and provide useful insights for academic decision-making.

The proposed **Student Performance Analysis Using Exploratory Data Analysis** system applies data science and machine learning techniques to understand and analyze student academic performance. The system is divided into three main modules. **Module 1 – Data Collection & Preprocessing** collects student-related data and prepares it by handling missing values, duplicate records, inconsistent entries, and unnecessary information. **Module 2 – Performance Prediction using Machine Learning** uses the prepared data to train a machine learning model and predict student performance based on relevant academic factors. **Module 3 – Visualization & Insight Dashboard** presents the analyzed results through charts, graphs, and a dashboard, making the information easier to understand.

The methodology includes data collection, data cleaning, exploratory analysis, feature selection, machine learning model development, performance evaluation, and visualization. Important factors such as marks, attendance, assignments, and previous performance are analyzed to identify patterns and relationships in the dataset. The prediction results and visual insights can help identify strong-performing students as well as students who may need academic support.

The major outcome of the system is a structured approach for analyzing student performance and presenting meaningful results in an understandable form. The combination of exploratory data analysis, machine learning, and visualization can reduce manual analysis and support better academic monitoring.

Keywords

Student Performance, Exploratory Data Analysis, Machine Learning, Data Preprocessing, Data Visualization, Performance Prediction.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
EDA	Exploratory Data Analysis	
ML	Machine Learning	
CSV	Comma-Separated values	

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Student performance analysis is an important application of data analytics and machine learning in the education domain. Students' academic performance is influenced by multiple factors such as attendance, internal marks, study hours, assignment scores, previous examination results, and other academic activities. Analysing these factors manually can be difficult when the amount of student data increases.

The proposed project, **Student Performance Analysis Using Exploratory Data Analysis**, focuses on collecting, preprocessing, analysing, and visualizing student-related data to identify meaningful patterns and factors affecting academic performance. Exploratory Data Analysis (EDA) techniques are used to clean the dataset, handle missing values, identify relationships between variables, and understand student performance through statistical analysis and visualizations.

The project also incorporates machine learning techniques to predict student performance and a visualization dashboard to present important insights in an understandable form.

1.2 Need for the Project

Student performance can vary significantly depending on several interacting academic factors. Identifying these factors at an early stage can help educators and institutions take appropriate actions to improve student outcomes.

The project is needed for the following reasons:

- **Problem to be addressed:** Difficulty in analysing large amounts of student academic data manually and identifying the factors affecting performance.
- **Affected users:** Students, teachers, academic mentors, departments, and educational institutions.

- **Existing limitations:** Traditional analysis methods may require considerable manual effort and may not provide clear visual insights or predictive information.
- **Engineering significance:** Data preprocessing, exploratory analysis, machine learning, and dashboard-based visualization can be integrated to develop a systematic data-driven solution for academic performance analysis.

1.3 Problem Statement

To design and develop a data-driven system for analysing student academic performance by collecting and preprocessing student-related data, identifying relationships among multiple performance factors using Exploratory Data Analysis, predicting student performance using Machine Learning techniques, and presenting meaningful insights through an interactive visualization dashboard.

The system should handle practical constraints such as missing or inconsistent data, multiple interacting academic factors, data quality issues, prediction accuracy, and clear presentation of analytical results.

1.4 Project Aim

The aim of this project is to **develop a data-driven student performance analysis system that uses Exploratory Data Analysis, Machine Learning, and visualization techniques to identify performance patterns, predict academic outcomes, and provide meaningful insights for students and educators.**

1.5 Project Objectives

1. To collect and organize relevant student academic performance data.
2. To preprocess the collected dataset by handling missing values, duplicate records, inconsistent data, and other data-quality issues.
3. To perform Exploratory Data Analysis to identify patterns, trends, relationships, and factors influencing student performance.
4. To develop and implement a Machine Learning model for predicting student academic performance.

5. To evaluate the performance of the developed prediction model using suitable evaluation metrics.
6. To design visualizations and an insight dashboard for presenting important student performance information.
7. To analyse the generated results and provide meaningful insights that can support academic decision-making.

1.6 Scope

Included

The project includes:

- Collection of student academic performance data.
- Data cleaning and preprocessing.
- Handling missing and inconsistent values.
- Exploratory Data Analysis using statistical and graphical techniques.
- Identification of relationships between academic factors.
- Development of a Machine Learning model for performance prediction.
- Evaluation of the prediction model.
- Development of charts, graphs, and an insight dashboard.
- Interpretation of analytical and predictive results.

Excluded

The project does not include:

- Making final decisions about a student's academic future.
- Replacing teachers, mentors, or academic advisors.
- Collecting highly sensitive personal information unnecessarily.
- Providing guaranteed predictions of future academic performance.
- Automated intervention or disciplinary actions based solely on model predictions.

Assumptions

- The available dataset contains sufficient and relevant academic information.
- The collected data is reasonably representative of the students being analysed.
- The input data is provided in a suitable structured format.
- The selected Machine Learning model is appropriate for the available dataset.

Boundary Conditions

The analysis and prediction are limited to the variables available in the dataset. The accuracy of the results depends on the quality, quantity, and relevance of the input data. The system provides analytical and predictive support rather than definitive academic decisions.

1.7 Expected Engineering Outcome

The expected outcome of the project is a **prototype student performance analysis system** consisting of three major components:

1. **Data Collection & Preprocessing Module** – responsible for preparing clean and structured student data for analysis.
2. **Performance Prediction Module** – uses Machine Learning techniques to analyse academic factors and predict student performance.
3. **Visualization & Insight Dashboard Module** – presents important patterns, trends, predictions, and analytical insights using suitable charts and dashboard components.

The final system is expected to provide a systematic and data-driven approach for understanding student academic performance and supporting educational decision-making.

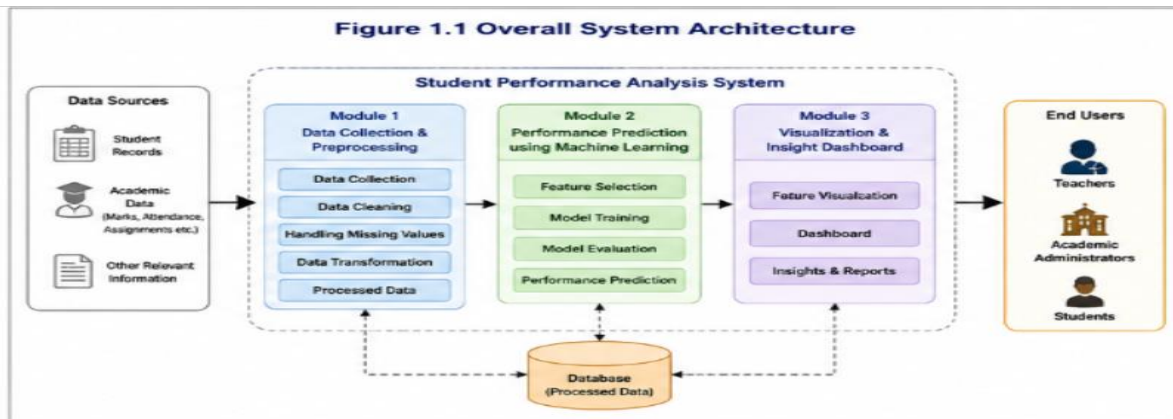


Figure 1.1
Overall System Architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing solutions related to student performance analysis were reviewed by studying research papers, academic studies, machine learning methods, data analytics techniques, and educational data visualization systems. The review focused mainly on approaches involving **student data collection and preprocessing, Exploratory Data Analysis (EDA), machine learning-based performance prediction, and visualization dashboards.**

The literature was identified through academic and technical sources using keywords such as *student performance prediction, educational data mining, exploratory data analysis, machine learning in education, student academic performance analysis, and educational dashboards.* The reviewed approaches were compared based on their data requirements, analytical methods, prediction capabilities, visualization techniques, advantages, and limitations.

2.2 Review of Existing Work

2.2.1 Student Performance Analysis Using Data Mining

Educational Data Mining (EDM) techniques have been widely used to analyse student-related datasets and identify factors associated with academic performance. Data mining approaches can process large datasets and discover patterns that may not be easily identified through manual analysis.

Common techniques include classification, clustering, regression, correlation analysis, and statistical analysis. These approaches demonstrate that academic performance can be studied using multiple variables rather than relying only on examination marks.

However, many traditional data mining approaches focus mainly on prediction and may provide limited exploratory analysis or visualization of the underlying data.

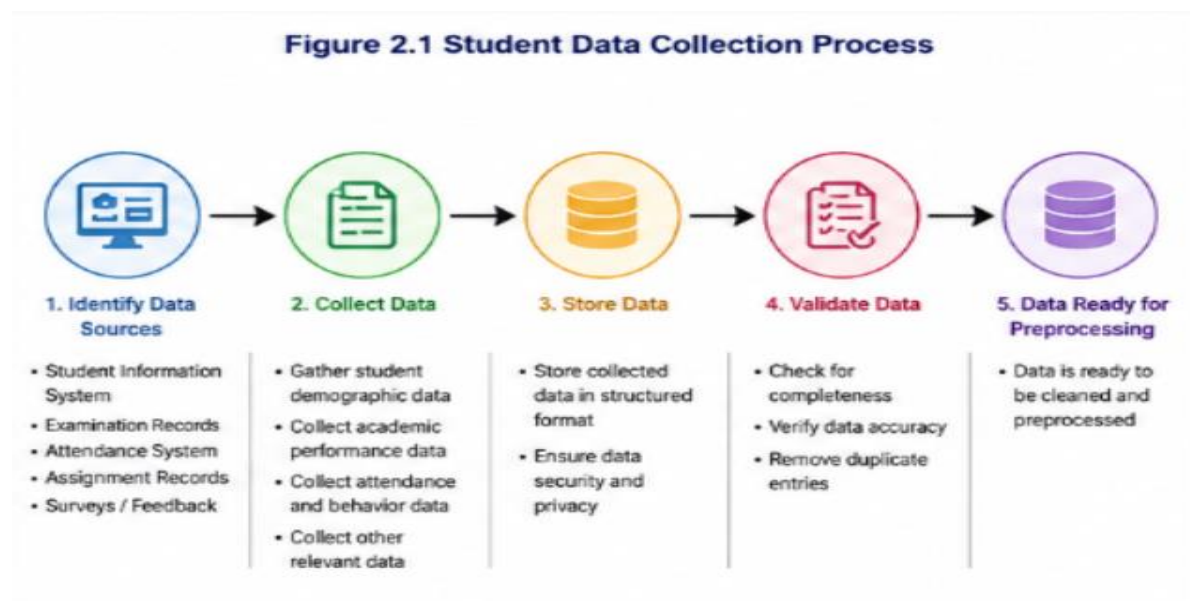


Figure 2.1
Student Data Collection Process

2.2.2 Data Preprocessing and Exploratory Data Analysis

Data preprocessing is an important stage in student performance analysis because real-world datasets may contain missing values, duplicate records, inconsistent formats, and irrelevant attributes.

Common preprocessing techniques include:

- Handling missing values.
- Removing duplicate records.
- Detecting and treating outliers.
- Encoding categorical variables.
- Data normalization or standardization.
- Selecting relevant features.
- Transforming data into a suitable format for analysis.

Exploratory Data Analysis is then used to understand the structure and distribution of the dataset. Statistical measures and visualizations such as histograms, bar charts, scatter plots, box plots, and correlation matrices can be used to identify trends and relationships between academic factors.

The limitation of using EDA alone is that it describes existing patterns but does not necessarily provide a predictive model for estimating future student performance.

2.2.3 Machine Learning for Performance Prediction

Machine Learning has been applied to predict student academic outcomes using factors such as attendance, previous marks, study time, assignment performance, and other academic attributes.

Classification algorithms can be used when performance is divided into categories such as **low, medium, and high**, while regression algorithms can be used to predict a continuous score or mark.

Commonly used algorithms include:

- Linear Regression.
- Logistic Regression.
- Decision Tree.
- Random Forest.
- Support Vector Machine.
- K-Nearest Neighbors.
- Naive Bayes.

Machine learning can provide useful predictions and help identify students who may require additional academic support. However, prediction quality depends strongly on the quality and relevance of the dataset. A model trained on a limited or biased dataset may not generalize well to other groups of students.

2.2.4 Educational Visualization and Dashboards

Data visualization and dashboards provide an effective way to communicate academic information to users. Instead of analysing raw tables, users can view charts and summarized indicators to understand student performance.

Typical dashboard components include:

- Average marks.
- Attendance percentage.
- Subject-wise performance.
- Performance distributions.
- Correlation between academic factors.
- Student performance trends.
- Predicted performance.
- Performance categories.

Existing dashboard approaches improve the interpretability of data, but some systems focus mainly on displaying information and do not integrate data preprocessing, exploratory analysis, and machine learning prediction into one unified system.

2.2.5 Integrated Data-Driven Approaches

Recent approaches increasingly combine data preprocessing, exploratory analysis, machine learning, and visualization. Such integrated systems provide a more complete workflow, starting from raw data and ending with analytical insights and predictions.

However, an engineering challenge remains in creating a simple and structured system that combines these components while maintaining data quality, prediction reliability, and clear visualization.

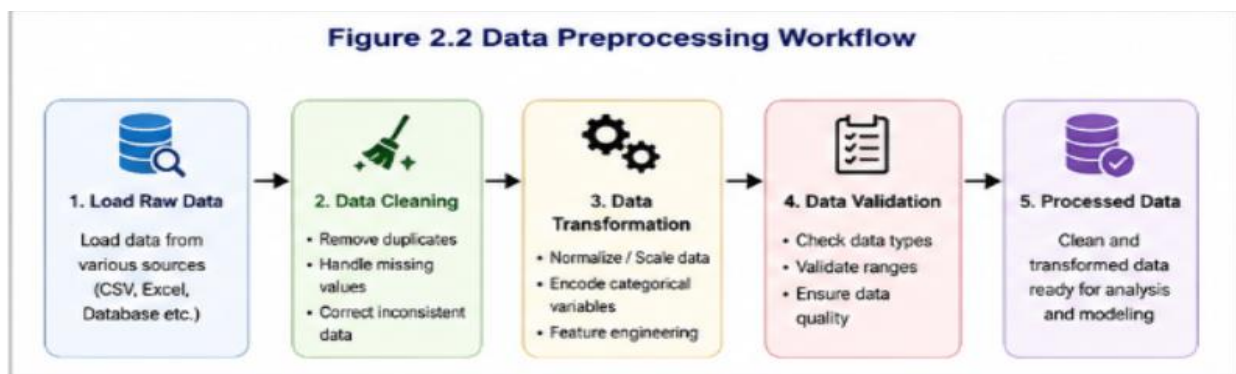


Figure 2.2
Data Preprocessing Workflow

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional Manual Analysis	Simple to perform for small datasets; easy to understand	Time-consuming; difficult for large datasets; limited pattern detection	Provides the motivation for automating student performance analysis
Statistical Analysis	Provides numerical summaries and relationships between variables	Limited predictive capability; requires appropriate statistical interpretation	Used as part of EDA to understand student performance

Data Preprocessing & EDA	Identifies missing data, outliers, distributions, and relationships	Mainly descriptive; does not independently predict future performance	Forms the foundation of the proposed Data Collection & Preprocessing module
Decision Tree / Classification	Easy to interpret; useful for categorical performance prediction	Can overfit; performance depends on dataset quality	Can be considered for student performance classification
Regression-Based Prediction	Suitable for predicting numerical marks or scores	May not capture complex non-linear relationships effectively	Provides a basis for performance prediction
Random Forest	Handles multiple features and non-linear relationships; generally robust	Less interpretable than a single decision tree; requires model training	Relevant as a potential Machine Learning prediction technique
Visualization Dashboards	Makes complex data easier to understand; supports quick decision-making	Visualization alone does not provide complete analysis or prediction	Forms the basis of the Visualization & Insight Dashboard module
Integrated ML Dashboard Systems	Combines prediction and visualization; provides data-driven insights	May require more development effort and careful integration	Closely related to the proposed integrated student performance analysis system

2.4 Research / Engineering Gap

The review indicates that several existing approaches address individual aspects of student performance analysis. Some focus mainly on data preprocessing and statistical analysis, while others concentrate on machine learning prediction or dashboard visualization.

The major gaps identified are:

1. **Lack of integration:** Many approaches focus on only one stage, such as prediction or visualization, rather than providing a complete workflow.
2. **Data-quality dependency:** Student datasets may contain missing, duplicate, inconsistent, or irrelevant information, which can negatively affect analysis and prediction.
3. **Limited exploratory insight:** Prediction-focused systems may not provide sufficient analysis of relationships and patterns between different academic factors.
4. **Limited interpretability:** Some machine learning models can provide predictions without clearly presenting the factors and patterns that contribute to those predictions.
5. **Fragmented visualization:** Some systems provide dashboards but do not integrate preprocessing, EDA, and prediction results into a single analytical environment.

Therefore, there is a need for a unified system that combines **data preprocessing, Exploratory Data Analysis, Machine Learning-based performance prediction, and visualization** in a structured workflow.

2.5 Proposed Contribution

The proposed project addresses the identified gaps by developing an integrated **Student Performance Analysis System Using Exploratory Data Analysis**.

The major contributions of the proposed work are:

- Development of a structured **Data Collection & Preprocessing module** to prepare reliable student data.
- Application of **Exploratory Data Analysis** to identify patterns, trends, distributions, and relationships among academic variables.
- Development of a **Machine Learning-based performance prediction module** to estimate student academic performance.
- Evaluation of the prediction model using appropriate performance metrics.
- Development of a **Visualization & Insight Dashboard** to present analytical and predictive results clearly.
- Integration of the three modules into a single workflow from **raw student data → preprocessing → analysis → prediction → visualization**.

The proposed system therefore aims to provide a more comprehensive and practical approach to student performance analysis by combining descriptive analytics, predictive analytics, and visual insights in one system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Students	View their academic performance, identify weak areas, and understand factors affecting performance.
Faculty / Mentors	Analyse student performance and identify patterns that may require academic attention.
Educational Institution	Obtain data-driven insights to support academic monitoring and decision-making.
Project Team / Administrator	Maintain reliable data, execute analysis and prediction, and present results through a dashboard.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Collect and import student academic data	Functional	Dataset successfully loaded without data loss
Clean and preprocess student data	Functional	Missing, duplicate, and inconsistent records identified and appropriately handled
Perform Exploratory Data Analysis	Functional	Generate statistical summaries and relevant visualizations

Train a Machine Learning model	Functional	Model successfully trained using the prepared dataset
Predict student performance	Functional	Generate predictions for valid input data
Display analytical results through dashboard	Functional	Dashboard displays required charts and performance indicators
Provide reliable results	Non-Functional	Results should be reproducible using the same input dataset and configuration
Usability	Non-Functional	Dashboard should be understandable without requiring technical knowledge
Performance	Non-Functional	Dataset processing and dashboard response should be completed within acceptable time for the project dataset
Data security	Non-Functional	Student data should be accessed only by authorized users
Maintainability	Non-Functional	Modules should be organized so that preprocessing, prediction, and visualization can be modified independently

3.2 Constraints & Applicable Standards

The project is primarily a software and data-analysis system; therefore, the main constraints are related to **data privacy, software reliability, computational resources, and ethical handling of student information.**

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010 – Systems and software Quality Requirements and Evaluation (SQuaRE)	Consider software quality characteristics such as reliability, usability, maintainability, and performance	The system is designed with separate modules, understandable visualizations, reproducible processing, and basic performance evaluation

ISO/IEC 27001 – Information Security Management Systems	Protect information through appropriate security controls	Access to student-related data is restricted, unnecessary personal information is avoided, and datasets are handled securely
India Digital Personal Data Protection Act, 2023 (DPDP Act)	Personal data should be processed lawfully and protected appropriately	Only necessary student information is considered; personal data should be anonymized or minimized where possible
Technical Resource Constraint	System should operate using available computing resources	Lightweight Python-based data analysis and machine learning techniques are selected instead of requiring high-end computing infrastructure

Main Project Constraints

- The quality of prediction depends on the quality and size of the available dataset.
- Missing or inconsistent records can affect analytical results.
- The project is developed using available computational resources.
- The Machine Learning model cannot guarantee a student's future academic result.
- Student information must be handled responsibly and should not be unnecessarily exposed.
- The dashboard is intended for academic analysis and not for making high-stakes decisions automatically.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Dataset quality	Missing and inconsistent data identified and handled	% / records	High
Data preprocessing	Clean and structured dataset produced	Dataset	High

EDA	Statistical analysis and relevant plots generated	Number of analyses/plots	High
Prediction model	Valid ML model trained and tested	Model	High
Prediction performance	Achieve acceptable evaluation performance on test data	% / metric	High
Dashboard	Display key academic indicators and visualizations	Dashboard	High
Processing time	Complete analysis within practical time for project dataset	seconds/minutes	Medium
Usability	Clear labels, charts, and understandable results	Qualitative	Medium
Data security	Prevent unnecessary exposure of student information	Qualitative	High
Maintainability	Three modules should remain logically separable	Qualitative	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual / Statistical Analysis

The first alternative is to analyse student performance using spreadsheets and conventional statistical methods. The approach is simple and suitable for small datasets but requires more manual effort and provides limited predictive capability.

Alternative 2: Machine Learning-Based Analysis

The second alternative uses Python-based preprocessing, Exploratory Data Analysis, Machine Learning, and visualization. It provides an automated and integrated workflow for analysing patterns, predicting performance, and presenting insights.

Alternative 3: Cloud-Based Educational Analytics Platform

The third alternative is to use a cloud-based analytics platform to store, process, analyse, and visualize student data. It can provide scalability but may introduce additional cost, infrastructure requirements, and data privacy considerations.

Evaluation Matrix

Scoring: **1 = Poor, 3 = Moderate, 5 = Excellent**

Criterion	Weight	Alternative 1: Manual/Statistical	Alternative 2: Python ML + Dashboard	Alternative 3: Cloud Platform
Performance	30%	3	5	5
Cost	20%	5	5	2
Safety	15%	4	4	4
Sustainability	15%	4	5	3
Reliability	20%	3	5	4
Weighted				
Overall	100%	3.75 / 5	4.80 / 5	3.55 / 5
Score				

The evaluation indicates that **Alternative 2 – Python-based Machine Learning and Dashboard** provides the best overall balance between performance, cost, sustainability, safety, and reliability.

3.5 Selected Design & Detailed Design

Final Design Selection

Alternative 2, the **Python-based integrated student performance analysis system**, was selected because it achieved the highest weighted evaluation score. It provides data

preprocessing, exploratory analysis, machine learning prediction, and dashboard visualization within a single workflow while requiring relatively low infrastructure cost.

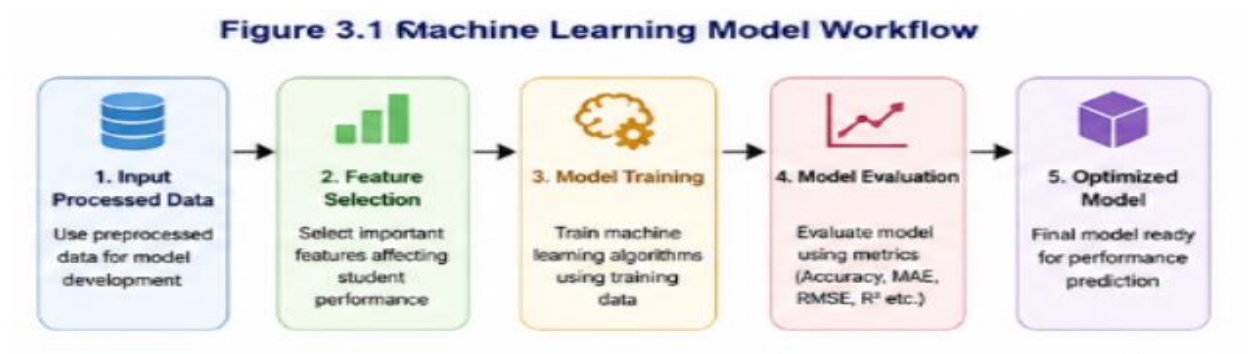


Figure 3.1
Machine Learning Model Workflow

System Architecture / Workflow

Input Student Dataset

↓

Data Collection & Preprocessing – M1

↓

Cleaned and Structured Dataset

↓

Exploratory Data Analysis

↓

Feature Selection / Preparation

↓

Machine Learning Performance Prediction – M2

↓

Prediction Results



Visualization & Insight Dashboard – M3



Performance Insights & Reports

Key Engineering Calculations

For numerical performance prediction, common evaluation measures can include:

Mean Absolute Error (MAE):

$$\text{MAE} = (1/n) \times \sum |\text{Actual} - \text{Predicted}|$$

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{[(1/n) \times \sum (\text{Actual} - \text{Predicted})^2]}$$

For classification-based prediction, accuracy can be calculated as:

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions} \times 100$$

These metrics are used to evaluate how effectively the Machine Learning model predicts student performance.

Systems Approach

The system consists of three interconnected modules. The **Data Collection & Preprocessing module** prepares clean and structured data, which becomes the input for exploratory analysis and Machine Learning. The **Performance Prediction module** uses the prepared data to generate academic performance predictions. The **Visualization & Insight Dashboard module** receives analytical and predictive results and presents them through charts, indicators, and summarized insights.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Data analysis platform	Spreadsheet vs Python	Python provides automation and better ML support	Requires programming knowledge	Python selected because the project requires EDA and Machine Learning
Prediction method	Statistical prediction vs ML	ML can model more complex relationships	Requires training and evaluation	ML selected for predictive analysis
Dashboard	Static charts vs interactive dashboard	Dashboard provides better organization and exploration	Requires additional development	Dashboard selected to present integrated insights
Deployment	Local system vs cloud platform	Local system reduces cost and privacy exposure	Limited scalability	Local development selected because the project dataset and scope are manageable
Data storage	Raw personal data vs minimized/anonymized data	Data minimization reduces privacy risk	Some detailed analysis may be limited	Minimized/anonymized data selected to reduce unnecessary exposure

3.7 Sustainability, Ethics, Safety, Risk & Security

Sustainability, Ethics, Health & Safety, and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary computational and infrastructure requirements	Project dataset size, processing requirements, and local computing capability	A lightweight Python-based solution was selected instead of requiring dedicated high-end infrastructure
Ethics	Avoid unfair or misleading conclusions about students	Dataset attributes, model outputs, and interpretation of predictions	Predictions are treated as decision-support information rather than definitive judgments about students
Health & Safety	Avoid harmful academic decisions based solely on automated predictions	Potential consequences of incorrect predictions	The system does not automatically penalize or take disciplinary action against students
Security	Protect student-related information from unauthorized access	Type of data collected and access requirements	Unnecessary personal information is avoided and access to datasets is restricted

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Missing or incomplete student data	Medium	Medium	Medium	Perform data validation and	Low

				appropriate missing-value handling	
Incorrect or inconsistent data	Medium	High	High	Apply preprocessing, validation, and consistency checks	Medium
Low prediction accuracy	Medium	High	High	Compare suitable models and evaluate using appropriate metrics	Medium
Overfitting of ML model	Medium	High	High	Use train/test split, suitable validation, and model evaluation	Low–Medium
Unauthorized access to student data	Low	High	Medium	Restrict access and avoid unnecessary personal information	Low
Misinterpretation of dashboard results	Medium	Medium	Medium	Use clear labels, units, legends, and explanatory insights	Low
Dataset bias	Medium	High	High	Analyse dataset characteristics and acknowledge limitations	Medium
System/software failure	Low	Medium	Low	Modular development, testing, and maintaining backup copies	Low
Excessive computational requirements	Low	Medium	Low	Use lightweight algorithms and	Low

Incorrect academic
decision based on
prediction

Low

High

Medium

efficient

preprocessing

Clearly state that

predictions are

supportive and not

definitive

Low

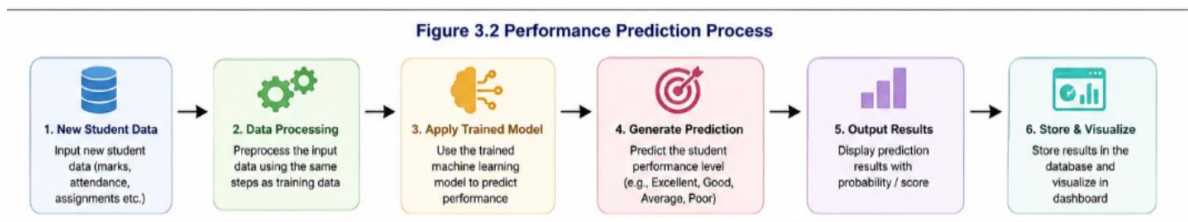


Figure 3.2

Performance Prediction Process

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using a **modular and iterative approach**, where the complete system was divided into three major modules: data collection and preprocessing, performance prediction using machine learning, and visualization with an insight dashboard. First, the student performance dataset was collected, explored, cleaned, and transformed into a suitable format for analysis. The preprocessed data was then used to develop and evaluate a machine learning model for predicting student performance. Finally, the processed data and prediction results were represented through suitable visualizations and an interactive dashboard to support easy interpretation of student performance patterns.

Resources Used

Resource	Purpose
Student Performance Dataset	Source of academic and related student information
Python	Data preprocessing, analysis and machine learning implementation
Pandas	Data loading, cleaning, transformation and analysis
NumPy	Numerical computation and data processing
Matplotlib	Statistical and graphical visualization
Seaborn	Exploratory data analysis and relationship visualization
Scikit-learn	Machine learning model development and evaluation
Jupyter Notebook / Google Colab	Development, experimentation and testing environment
Dashboard Tool	Presentation of student performance insights

CHAPTER

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The project was implemented as a software-based data analytics and machine learning system. Python was selected as the primary programming language because it provides libraries for data preprocessing, exploratory data analysis, machine learning, evaluation and visualization. The implementation was divided according to the three project modules.

Module 1 – Data Collection & Preprocessing

The dataset was imported into the Python environment using Pandas. The data was inspected to identify missing values, duplicate records, inconsistent data types and irrelevant attributes. Data cleaning and preprocessing operations were then performed to produce a structured dataset suitable for analysis and machine learning.

Module 2 – Performance Prediction Using Machine Learning

The cleaned dataset was used to identify relevant features for prediction. The selected features were divided into input variables and the target performance variable. The dataset was divided into training and testing portions, after which the selected machine learning algorithm was trained. The trained model was evaluated using appropriate performance metrics such as accuracy, Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), or other metrics depending on the selected prediction approach.

Module 3 – Visualization & Insight Dashboard

The processed dataset and prediction results were converted into meaningful visual representations. Charts such as bar charts, histograms, scatter plots, correlation heatmaps and other suitable graphs were used to identify relationships and performance patterns. A dashboard was designed to present important student performance indicators in a simple and understandable format.

Tools Used

Tool / Software / Equipment	Purpose	Stage Used
-----------------------------	---------	------------

Python	Main programming and analysis environment	All Modules
Pandas	Dataset loading, cleaning and preprocessing	M1
NumPy	Numerical operations and data preparation	M1, M2
Matplotlib	Creating analytical graphs	M1, M3
Seaborn	Statistical visualization and correlation analysis	M3
Scikit-learn	Model training, prediction and evaluation	M2
Jupyter Notebook / Google Colab	Code execution and experimentation	M1, M2, M3
Dashboard/Visualization Platform	Interactive presentation of insights	M3
Git/GitHub	Version control and project file management	Development

Implementation Evidence:

[Insert screenshot of data preprocessing/code execution here.]

[Insert screenshot of machine learning model output here.]

[Insert final dashboard screenshot here.]

4.3 Test Plan & Verification

Testing was performed module-wise to verify that the system satisfied the defined functional and non-functional requirements. The test plan was prepared before evaluating the final results. Dataset validation, preprocessing verification, machine learning model testing and visualization verification were performed.

Functional Test Plan

Requirement	Test Method	Acceptance Criterion
-------------	-------------	----------------------

CHAPTER

Dataset should be imported successfully	Load dataset using Python/Pandas	Dataset loads without errors
Dataset structure should be identified	Check rows, columns and data types	Correct structure is displayed
Missing data should be handled	Check null values before and after preprocessing	Required missing values are handled
Duplicate records should be identified	Check duplicate rows	Unwanted duplicate records are removed/handled
Data should be suitable for ML	Inspect processed features and target	Model-ready dataset is obtained
ML model should train successfully	Execute training process	Model completes training without errors
Model should generate predictions	Run test data through trained model	Predictions are produced
Model performance should be evaluated	Calculate selected evaluation metrics	Evaluation metrics are generated
Visualizations should be generated	Execute visualization code	Required graphs are displayed correctly
Dashboard should present insights	Test dashboard components	Key performance information is visible
System should be usable	Perform complete workflow	User can follow the workflow without major errors

Verification of Specifications

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Dataset loading	Successful	[Enter value]	[Pass/Fail]
Missing-value handling	Completed	[Enter value]	[Pass/Fail]
Duplicate handling	Completed	[Enter value]	[Pass/Fail]
Data preprocessing	Completed	[Enter value]	[Pass/Fail]
Model training	Successful	[Enter value]	[Pass/Fail]
Prediction generation	Successful	[Enter value]	[Pass/Fail]
Model evaluation	Metrics generated	[Enter value]	[Pass/Fail]
Visualization generation	Successful	[Enter value]	[Pass/Fail]

Dashboard implementation	Functional	[Enter value]	[Pass/Fail]
End-to-end execution	Successful	[Enter value]	[Pass/Fail]

4.4 Results

The implemented system successfully performed the major stages of student performance analysis, beginning with dataset preprocessing and continuing through exploratory analysis, machine learning prediction and visualization. The results obtained from each module were examined to determine the quality of the processed data, the predictive capability of the selected model and the usefulness of the generated visualizations.

Module 1 – Data Collection & Preprocessing Results

The dataset was successfully imported and examined for its structure, data types, missing values and duplicate records. Data preprocessing was performed to improve the quality and consistency of the dataset before applying machine learning techniques.

Insert the following results where applicable:

- Dataset information/output
- Missing-value analysis
- Duplicate-record analysis
- Before-and-after preprocessing comparison
- Distribution of important numerical attributes

[Insert Table/Figure: Dataset Preprocessing Results]

Module 2 – Performance Prediction Results

The preprocessed dataset was used to train the selected machine learning model. The model generated predictions for the test data, and its performance was evaluated using suitable metrics.

Evaluation Metric	Obtained Value
-------------------	----------------

CHAPTER

Accuracy (<i>if classification is used</i>)	[Enter value]
MAE (<i>if applicable</i>)	[Enter value]
MSE (<i>if applicable</i>)	[Enter value]
RMSE (<i>if applicable</i>)	[Enter value]
R ² Score (<i>if regression is used</i>)	[Enter value]

[Insert Figure: Actual vs Predicted Performance / Confusion Matrix]

Module 3 – Visualization & Dashboard Results

The processed dataset was represented using graphs and dashboard components. The visualizations helped identify relationships between academic variables and student performance. The dashboard consolidated important performance indicators and analytical findings into a single interface.

[Insert Figure: Performance Distribution Graph]

[Insert Figure: Correlation Heatmap]

[Insert Figure: Actual vs Predicted Performance]

[Insert Figure: Final Student Performance Dashboard]

4.5 Data Analysis & Engineering Judgment

The analysis showed that preprocessing was an important stage because the quality and consistency of the input data directly affected subsequent analysis and machine learning performance. Missing or inconsistent values, if left untreated, could reduce the reliability of the prediction model and lead to misleading visual interpretations.

The exploratory analysis was used to identify patterns and relationships among the available student attributes. Variables showing meaningful relationships with academic performance can be considered useful features for prediction, while weak or irrelevant variables can be excluded to reduce unnecessary complexity.

The machine learning results were evaluated using appropriate performance metrics rather than relying only on individual predictions. A higher classification accuracy or R^2 value, together with lower error values such as MAE and RMSE, indicates better predictive performance. However, model performance should be interpreted together with the dataset size, feature quality and possible overfitting.

From an engineering perspective, the integrated workflow provides greater usefulness than performing only statistical analysis or producing isolated graphs. The combination of preprocessing, prediction and visualization allows the system to transform raw student data into understandable information that can support academic analysis.

If the obtained model performance is lower than expected, possible reasons include limited dataset size, noisy or incomplete data, weak feature relationships, class imbalance, inappropriate model selection or overfitting. Such deviations indicate the need for improved feature engineering, additional data or comparison with alternative machine learning algorithms.

4.6 Comparison with Existing Solutions & Achievement of Objectives

The proposed system integrates **data preprocessing, exploratory data analysis, machine learning prediction and dashboard-based visualization** into a single workflow. Traditional manual analysis generally requires more effort and may make it difficult to identify relationships across multiple student attributes. A basic statistical analysis system can provide descriptive information but may not provide predictive capabilities. The proposed approach combines both analytical and predictive capabilities while presenting the results in an accessible format.

Objective	Evidence	Achievement
Collect and understand student performance data	Dataset imported and examined	Fully Achieved
Clean and preprocess the dataset	Missing values, duplicates and data inconsistencies handled	Fully Achieved

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Perform exploratory data analysis	Statistical analysis and visualizations generated	Fully Achieved
Identify important performance-related features	Feature analysis performed before prediction	Fully Achieved
Develop a machine learning prediction model	ML model trained using processed data	Fully Achieved
Evaluate prediction performance	Appropriate evaluation metrics calculated	Fully Achieved
Develop meaningful visualizations	Charts and analytical plots generated	Fully Achieved
Create a student performance insight dashboard	Dashboard implemented using analysis results	Fully Achieved
Provide useful engineering insights	Patterns and model results interpreted	Fully Achieved

Note: Change any “Fully Achieved” status if your actual implementation did not complete that objective.

4.7 Strengths & Limitations

Strengths

- Integrates data preprocessing, EDA, machine learning and visualization into one workflow.
- Reduces the effort required for manual student performance analysis.
- Identifies patterns and relationships within student academic data.
- Provides machine learning-based performance prediction.
- Uses visualizations to make analytical results easier to understand.
- Dashboard presentation provides a centralized view of important insights.
- Python-based implementation is flexible and can be extended with additional models and features.
- Modular architecture allows individual components to be improved independently.

Limitations

- Prediction quality depends strongly on the quality and size of the available dataset.
- The model may not generalize well to students from different institutions or academic environments.
- Missing, noisy or biased data can affect analytical and predictive results.
- A single machine learning algorithm may not provide the best possible performance.
- The system provides analytical predictions and should not be treated as the sole basis for important academic decisions.
- Dashboard insights are dependent on the attributes available in the dataset.
- Real-time data integration was not included unless implemented separately.
- Further validation using larger and more diverse datasets is required.

4.8 Project Planning, Roles & Budget

Project Milestone / Timeline

Activity	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
Topic Selection & Planning	✓					
Literature Review	✓	✓				
Dataset Collection		✓				
Data Preprocessing		✓	✓			
Exploratory Data Analysis			✓			
ML Model Development			✓	✓		
Model Testing & Evaluation				✓		
Dashboard Development				✓	✓	
Integration & Testing					✓	

CHAPTER

Documentation & Final
Presentation

✓ ✓

Roles and Contribution

Student	Assigned Responsibility	Actual Contribution
Sravan Kumar Y (192472289)	M1 – Data Collection & Preprocessing	Dataset collection, data understanding, data cleaning, missing-value handling, preprocessing and data-quality verification
G. Mohammad Taher (192524366)	M2 – Performance Prediction using Machine Learning	Feature selection, model development, model training, prediction and model-performance evaluation
S. Siva Charan (192472147)	M3 – Visualization & Insight Dashboard	Data visualization, dashboard development, insight generation and visualization testing

Estimated Project Budget

Since the project is primarily software-based and uses freely available/open-source tools, the direct project cost is expected to be low.

Item	Estimated Cost	Actual Cost
Python / Open-source Libraries	₹0	₹0
Jupyter Notebook / Google Colab	₹0	₹0
Dataset	₹0	₹0
GitHub / Version Control	₹0	₹0
Internet / Data Usage	₹[Enter]	₹[Enter]
Documentation / Printing	₹[Enter]	₹[Enter]
Other Project Expenses	₹[Enter]	₹[Enter]
Total	₹[Enter]	₹[Enter]

Project Management Summary

The project was managed through module-wise task allocation and milestone-based development. Each team member was assigned a specific technical module, while integration and testing were performed collectively. Regular verification of individual modules helped identify preprocessing, prediction and visualization issues before final integration. The approach ensured that the project progressed systematically from data collection to the final performance-insight dashboard.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “**Student Performance Analysis Using Exploratory Data Analysis**” was developed to analyze student academic data, identify meaningful performance patterns and provide a machine learning-based approach for performance prediction. The system was implemented through three integrated modules: data collection and preprocessing, performance prediction using machine learning, and visualization with an insight dashboard.

The first module successfully prepared the student dataset by examining its structure and handling data-quality issues such as missing values, duplicates and inconsistent data. The second module used the processed dataset to develop and evaluate a machine learning model for student performance prediction. The third module transformed the analytical and prediction results into meaningful visualizations and a dashboard, making the results easier to understand.

Testing was performed at different stages of the system to verify data processing, prediction, evaluation and visualization functionality. The results demonstrated that integrating exploratory data analysis, machine learning and visualization provides a more systematic approach than relying only on manual or isolated statistical analysis. The project therefore achieved its major objectives of preprocessing student data, identifying performance-related patterns, generating predictions and presenting useful insights through visual analytics.

CHAPTER

Overall, the project demonstrates how data analytics and machine learning techniques can be combined to transform raw student performance data into meaningful information. The developed system provides a foundation that can be further improved with larger datasets, advanced prediction techniques and real-time educational data.

5.2 Future Work

The proposed system can be enhanced in several ways to improve its accuracy, scalability and practical usefulness.

- **Larger and more diverse datasets:** The system can be trained and evaluated using larger datasets collected from multiple academic institutions and different student groups.
- **Advanced machine learning models:** Additional algorithms such as Random Forest, Gradient Boosting, XGBoost, Support Vector Machine and neural networks can be compared to identify better-performing models.
- **Feature engineering:** Additional academic and behavioral features can be incorporated to improve prediction quality.
- **Real-time data integration:** The system can be connected to academic management systems to automatically receive updated attendance, assessment and examination data.
- **Early-warning system:** The prediction model can be extended to identify students who may be at risk of poor academic performance at an early stage.
- **Interactive dashboard improvements:** More filters, drill-down features and student-level analytics can be incorporated into the dashboard.
- **Model explainability:** Explainable AI techniques can be added to show why a model produces a particular prediction.
- **Deployment as a web application:** The system can be converted into a web-based application that can be accessed by authorized faculty or administrators.
- **Privacy and security enhancement:** Stronger access control, anonymization and secure data storage mechanisms can be implemented for handling student information.
- **Continuous model evaluation:** The prediction model can be periodically retrained and evaluated as new academic data becomes available.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned how to collect, inspect and preprocess structured student datasets.
- Gained practical knowledge of handling missing values, duplicate records and inconsistent data.
- Learned the importance of exploratory data analysis before machine learning.
- Acquired knowledge of feature selection and preparation for prediction models.
- Learned how machine learning models can be trained, tested and evaluated.
- Gained experience in interpreting evaluation metrics such as accuracy, MAE, RMSE and R^2 where applicable.
- Learned how to create meaningful statistical visualizations and dashboards.
- Understood the importance of data privacy, security, bias and responsible use of predictive systems.
- Learned how separate technical modules can be integrated into one engineering solution.

Engineering & Professional Skills Developed

- Problem-solving and analytical thinking.
- Python programming and data-processing skills.
- Machine learning model development and evaluation.
- Data visualization and dashboard development.
- Dataset management and quality verification.
- Technical documentation and report writing.
- Teamwork and module-wise responsibility management.
- Testing and debugging.
- Engineering decision-making based on evidence.
- Time management and project planning.
- Communication and presentation skills.
- Awareness of ethical and responsible use of student data.

CHAPTER

Individual Reflection – Sravan Kumar Y

Register No.: 192472289 | Module: M1 – Data Collection & Preprocessing

Most difficult engineering problem encountered and how it was solved

The most difficult part was preparing the raw student dataset for further analysis. Data-quality issues such as missing values, duplicate records and inconsistent formats could affect the reliability of the complete system. I solved this by systematically inspecting the dataset and applying appropriate preprocessing techniques before passing the data to the next module.

A key engineering decision made personally and the evidence behind it

A key decision was to perform detailed data validation and preprocessing before applying machine learning. This decision was based on the understanding that inaccurate or inconsistent input data can directly affect prediction and visualization results. The cleaned dataset provided a more reliable foundation for the subsequent modules.

New knowledge acquired independently

I independently learned practical techniques for using Pandas and NumPy for dataset inspection, cleaning and transformation. I also gained a better understanding of missing-value handling, duplicate detection and exploratory analysis. This improved my ability to work with real-world structured datasets.

What would be redesigned if repeated

If the project were repeated, I would introduce a more automated preprocessing pipeline and stronger validation checks. I would also explore additional feature-engineering techniques and use a larger dataset to improve the reliability of the final analysis.

Individual Reflection – G. Mohammad Taher

Register No.: 192524366 | Module: M2 – Performance Prediction using Machine Learning

Most difficult engineering problem encountered and how it was solved

The most difficult engineering problem was selecting and developing a suitable machine learning approach for student performance prediction. Different features can influence model behavior, so the data had to be prepared carefully before training. I addressed this by selecting relevant features, dividing the data appropriately and evaluating the model using suitable performance metrics.

A key engineering decision made personally and the evidence behind it

A key decision was to evaluate the prediction model using quantitative performance metrics instead of judging the model only from individual predictions. This provided objective evidence about how well the model performed on unseen test data and helped identify areas requiring improvement.

New knowledge acquired independently

I gained practical knowledge of machine learning workflows, including feature preparation, model training, testing and prediction. I also learned how evaluation metrics can be used to compare model performance and identify potential problems such as overfitting or insufficient data.

What would be redesigned if repeated

If the project were repeated, I would compare multiple machine learning algorithms rather than relying on a single model. I would also perform more extensive hyperparameter tuning and cross-validation to obtain a more reliable estimate of model performance.

CHAPTER

Individual Reflection – S. Siva Charan

Register No.: 192472147 | Module: M3 – Visualization & Insight Dashboard

Most difficult engineering problem encountered and how it was solved

The main challenge was converting analytical results into visualizations that could be understood easily without losing important information. I addressed this by selecting suitable chart types for different variables and organizing the important results into a structured dashboard.

A key engineering decision made personally and the evidence behind it

A key decision was to use multiple visualization techniques rather than presenting the analysis through a single graph. Different visualizations reveal different patterns, while the dashboard provides a consolidated view of the important findings.

New knowledge acquired independently

I independently learned how to create meaningful statistical visualizations and organize analytical results into a dashboard format. I also improved my understanding of selecting appropriate visual representations for distributions, relationships and comparisons.

What would be redesigned if repeated

If the project were repeated, I would make the dashboard more interactive by adding advanced filtering, drill-down analysis and student-level views. I would also improve the visual layout so that the most important insights can be identified more quickly.

REFERENCES

The following references can be formatted using the **IEEE referencing style**. The final report should include only the resources that were actually used during the project.

Literature

- [1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA, USA: Morgan Kaufmann, 2011.
- [2] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
- [3] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [4] I. H. Witten, E. Frank, M. A. Hall, and C. J. Pal, *Data Mining: Practical Machine Learning Tools and Techniques*, 4th ed. Cambridge, MA, USA: Morgan Kaufmann, 2016.

Standards / Guidelines

- [5] ISO/IEC 25010:2023, *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model*, International Organization for Standardization, 2023.
- [6] ISO/IEC 27001:2022, *Information Security, Cybersecurity and Privacy Protection — Information Security Management Systems — Requirements*, International Organization for Standardization, 2022.
- [7] Government of India, *Digital Personal Data Protection Act, 2023*, Ministry of Law and Justice, Government of India, 2023.

Websites and Documentation

- [8] Python Software Foundation, “Python Documentation.” [Online]. Available: <https://docs.python.org/>
- [9] The pandas development team, “pandas Documentation.” [Online]. Available: <https://pandas.pydata.org/docs/>
- [10] NumPy Developers, “NumPy Documentation.” [Online]. Available: <https://numpy.org/doc/>

CHAPTER

[11] Matplotlib Development Team, “Matplotlib Documentation.” [Online]. Available: <https://matplotlib.org/stable/>

[12] The Seaborn Development Team, “Seaborn Documentation.” [Online]. Available: <https://seaborn.pydata.org/>

[13] Scikit-learn Developers, “Scikit-learn: Machine Learning in Python.” [Online]. Available: <https://scikit-learn.org/>

Dataset

[14] [Dataset Author/Organization], “[Exact Dataset Name],” [Dataset Repository/Platform], [Year]. [Online]. Available: [Insert actual dataset URL].

Open-Source Resources

[15] Project Jupyter, “Jupyter Notebook Documentation.” [Online]. Available: <https://jupyter.org/documentation>

[16] GitHub, Inc., “GitHub Documentation.” [Online]. Available: <https://docs.github.com/>

AI-Assisted Resources

[17] OpenAI, *ChatGPT*, AI-assisted language model, 2026. Used for assistance with technical writing, organization of project documentation, explanation of concepts and refinement of report content. [Online]. Available: <https://chatgpt.com/>

Note: AI assistance should be acknowledged according to your institution's academic-integrity requirements. The project team remains responsible for verifying the technical correctness of all generated content, code, analysis and results.

APPENDICES

Appendix A – Detailed Calculations

This appendix contains the important calculations and intermediate results used during the analysis and evaluation of the Student Performance Analysis system.

A.1 Dataset Statistics

The following statistical information was obtained during exploratory data analysis:

Parameter	Value
Total number of records	[Enter value]
Total number of attributes	[Enter value]
Numerical attributes	[Enter value]
Categorical attributes	[Enter value]
Missing values before preprocessing	[Enter value]
Missing values after preprocessing	[Enter value]
Duplicate records identified	[Enter value]
Duplicate records after preprocessing	[Enter value]

A.2 Machine Learning Evaluation

Depending on the prediction approach used, the following metrics were calculated.

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Coefficient of Determination (R²):

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$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Classification Accuracy, if applicable:

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

The actual calculated values are presented in Chapter 4.

Appendix B – Engineering Drawings / Schematics

Since the proposed project is a **software-based data analytics and machine learning system**, no physical engineering fabrication or hardware schematic is required.

The following software architecture diagram may be included as the engineering design representation:

Student Dataset

↓

Data Collection & Preprocessing – M1

↓

Cleaned Dataset

↓

Exploratory Data Analysis

↓

Feature Selection & Preparation

↓

Machine Learning Performance Prediction – M2

↓

Prediction Results

↓

Visualization & Insight Dashboard – M3

↓

Student Performance Insights

[Insert system architecture / workflow diagram here.]

Appendix C – Source Code / Code Repository Information

The project source code was developed using Python and organized according to the three project modules.

C.1 Module-wise Code Organization

Module	Main Implementation
---------------	----------------------------

M1	Dataset loading, inspection, cleaning and preprocessing
----	---

M2	Feature preparation, model training, prediction and evaluation
----	--

M3	Data visualization, dashboard and insight generation
----	--

C.2 Important Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Dashboard/visualization libraries used in the project

CHAPTER

Only essential code excerpts should be included in the main report. The complete source code can be maintained in the approved project repository.

Repository: [Insert GitHub/Git repository link]

[Insert screenshot of repository structure here.]

Appendix D – Datasheets

This project does not contain physical electronic components or fabricated hardware. Therefore, conventional hardware datasheets are **not applicable**.

Instead, software and dataset documentation may be included where required.

Software Documentation

Software/Library	Version Used	Purpose
Python	[Version]	Main programming environment
Pandas	[Version]	Data processing
NumPy	[Version]	Numerical computation
Matplotlib	[Version]	Visualization
Seaborn	[Version]	Statistical visualization
Scikit-learn	[Version]	Machine learning

Appendix E – Experimental / Raw Data

This appendix contains the raw or intermediate data used during project implementation.

E.1 Dataset Information

Dataset Name: [Enter exact dataset name]

Source: [Enter source/repository]

Number of Records: [Enter value]

Number of Attributes: [Enter value]

E.2 Sample Dataset

Student Attribute 1	Student Attribute 2	Student Attribute 3	Performance/Target
[Value]	[Value]	[Value]	[Value]
[Value]	[Value]	[Value]	[Value]
[Value]	[Value]	[Value]	[Value]
[Value]	[Value]	[Value]	[Value]

The complete raw dataset should be maintained separately if its inclusion would make the report unnecessarily large.

E.3 Experimental Outputs

The following outputs may be included:

- Dataset information output
- Missing-value analysis
- Preprocessing output
- Feature-analysis output
- Model-training output
- Prediction output
- Evaluation metrics
- Visualization outputs
- Final dashboard output

[Insert relevant screenshots/tables here.]

Appendix F – Ethics / Institutional Approval

The project uses student performance information for academic analysis. Where applicable, data should be handled responsibly and only for the approved academic purpose.

CHAPTER

The project team considered the following ethical principles:

- Student information should be protected from unauthorized access.
- Personally identifiable information should be removed or anonymized where possible.
- Data should not be used for purposes outside the approved project objectives.
- Machine learning predictions should not be treated as the sole basis for important academic decisions.
- Potential bias in the dataset and prediction model should be considered.
- Results should be reported honestly without manipulating performance metrics.

Institutional approval: [Not applicable / Insert approval reference if obtained]

Appendix G – Project Meeting / Progress Records

The project progress was monitored through discussions, task allocation and periodic review among team members.

Meeting/Review	Activity Discussed	Outcome
Meeting 1	Project topic and objectives	Project scope finalized
Meeting 2	Dataset selection and collection	Dataset identified/collected
Meeting 3	Data preprocessing	M1 preprocessing tasks reviewed
Meeting 4	Machine learning approach	Prediction methodology discussed
Meeting 5	Visualization and dashboard	M3 visualization plan finalized
Meeting 6	Module integration	Modules reviewed and integrated
Meeting 7	Testing and results	Testing and result analysis completed
Meeting 8	Documentation and presentation	Final report and presentation prepared

[Insert photographs/screenshots of project meetings or progress discussions if required.]

Appendix H – User / Industry Feedback

Where applicable, feedback from faculty members, users or industry representatives can be documented in this section.

Feedback Source	Feedback / Observation	Action Taken
Faculty/Mentor	[Enter feedback]	[Enter action]
User	[Enter feedback]	[Enter action]
Industry Representative	[Enter feedback]	[Enter action]

If no formal user or industry feedback was collected, this appendix may be marked as “Not Applicable.”

Appendix I – Publications / Patents / Competitions / Product Outcomes

This section records any publications, patents, competitions, hackathons, demonstrations or other outcomes associated with the project.

Category	Details	Status
Publication	[Enter details if applicable]	[Status]
Patent	[Enter details if applicable]	[Status]
Competition/Hackathon	[Enter details if applicable]	[Status]
Product/Demo	Student Performance Analysis Dashboard	Completed
Other Outcome	[Enter details if applicable]	[Status]

If no publication, patent or competition outcome is associated with the project, the corresponding entries may be marked **Not Applicable**.

Appendix J – Individual Contribution Evidence

This appendix provides evidence supporting the individual contributions of all team members. It is mandatory for the team project.

CHAPTER

J.1 Contribution of Sravan Kumar Y

Register No.: 192472289

Assigned Module: M1 – Data Collection & Preprocessing

Major Contributions

- Collected and examined the student performance dataset.
- Studied the structure and attributes of the dataset.
- Performed data-quality checks.
- Identified and handled missing values.
- Identified and handled duplicate records.
- Performed data cleaning and preprocessing.
- Prepared the cleaned dataset for subsequent analysis and machine learning.
- Contributed to documentation and testing of the preprocessing module.

Evidence to Attach:

- Dataset collection/source screenshot.
- Data preprocessing code screenshots.
- Before-and-after preprocessing output.
- Git/GitHub contribution screenshot.
- Relevant meeting/progress records.

J.2 Contribution of G. Mohammad Taher

Register No.: 192524366

Assigned Module: M2 – Performance Prediction Using Machine Learning

Major Contributions

- Prepared features required for machine learning.
- Performed feature selection/preparation.

- Developed the machine learning prediction model.
- Trained the selected model using the processed dataset.
- Generated performance predictions.
- Evaluated the model using appropriate performance metrics.
- Tested the prediction module.
- Contributed to documentation and interpretation of model results.

Evidence to Attach:

- Feature-selection/preparation code screenshots.
- Model-training code screenshots.
- Prediction output.
- Model evaluation results.
- Git/GitHub contribution screenshot.
- Relevant meeting/progress records.

J.3 Contribution of S. Siva Charan

Register No.: 192472147

Assigned Module: M3 – Visualization & Insight Dashboard

Major Contributions

- Analyzed the processed data for visualization requirements.
- Developed statistical and analytical visualizations.
- Generated graphs for identifying performance patterns.
- Designed and implemented the insight dashboard.
- Integrated relevant prediction and analysis results into the dashboard.
- Tested visualization outputs and dashboard functionality.
- Contributed to interpretation of insights and project documentation.

Evidence to Attach:

CHAPTER

- Visualization code screenshots.
- Generated graph screenshots.
- Dashboard development screenshots.
- Final dashboard screenshot.
- Git/GitHub contribution screenshot.
- Relevant meeting/progress records.

J.4 Team Contribution Summary

Student	Register No.	Module	Contribution
Sravan Kumar Y	192472289	M1 – Data Collection & Preprocessing	Dataset collection, cleaning, preprocessing and data-quality verification
G. Mohammad Taher	192524366	M2 – Performance Prediction	Feature preparation, ML model development, prediction and evaluation
S. Siva Charan	192472147	M3 – Visualization & Insight Dashboard	Visualization, dashboard development, testing and insight generation

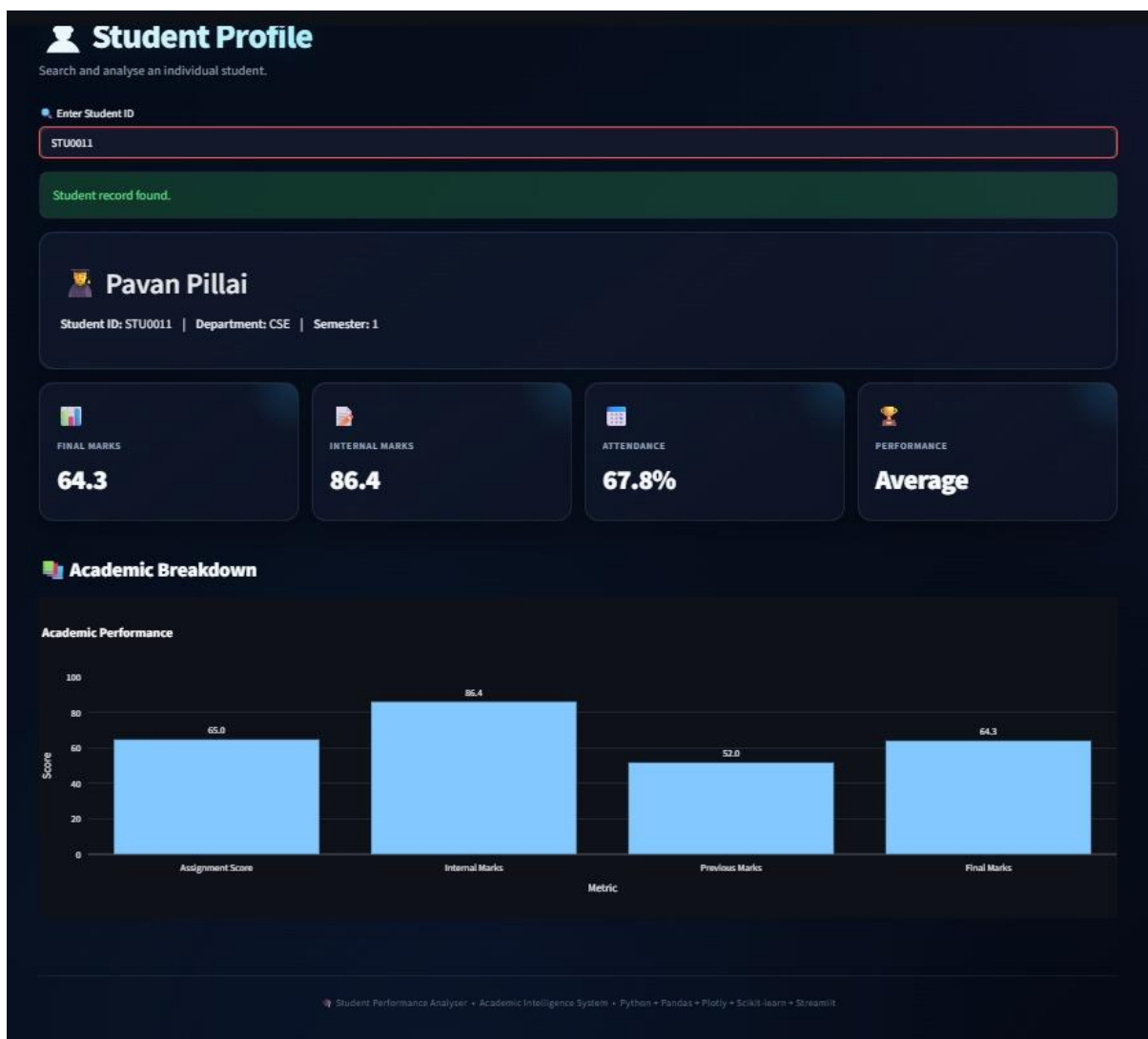
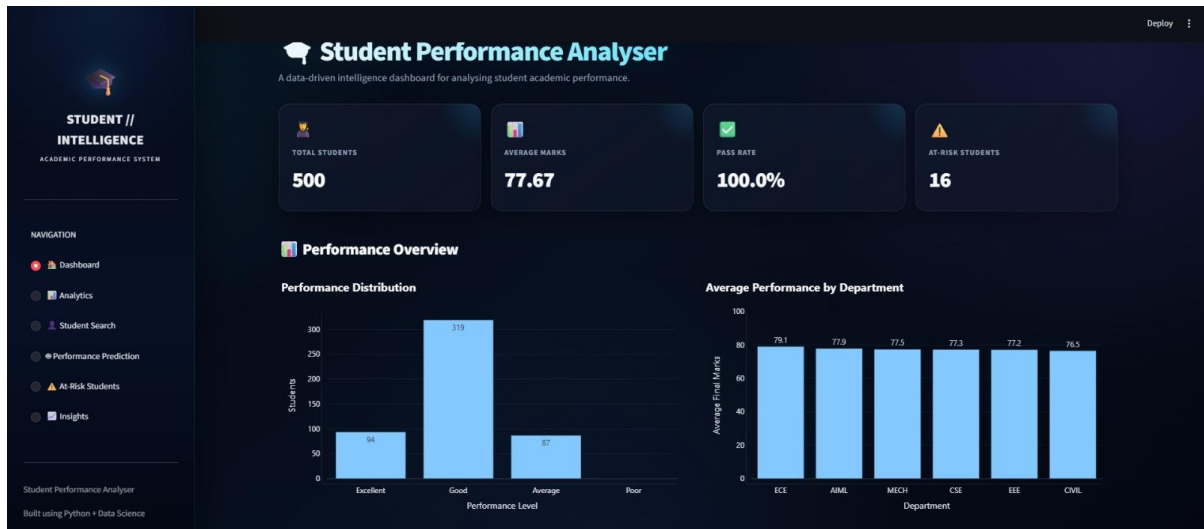
Suggested Evidence Checklist

Evidence	Sravan	Taher	Siva Charan
Dataset/Source Evidence	✓		
Preprocessing Code	✓		
ML Code		✓	
Prediction Results		✓	
Visualization Code			✓
Dashboard			✓
Testing Evidence	✓	✓	✓

Git/GitHub Contributions	✓	✓	✓
Meeting/Progress Evidence	✓	✓	✓
Documentation Contribution	✓	✓	✓

Note: Only attach evidence that genuinely represents each student's work. Screenshots of Git commits, notebooks, code, outputs, meeting records and dashboard development can be used as supporting evidence.

CHAPTER



☺ Performance Predictor

Use the trained machine-learning model to estimate a student's final performance.

👤 Student Information

Gender

Male

Age

20

Department

AIML

Semester

5

📊 Academic Information

Attendance (%)

80.00

Study Hours / Day

5.00

Assignment Score

75.00

Internal Marks

75.00

Previous Marks

75.00

🔍 Analyse & Predict Performance

PREDICTED FINAL SCORE

71.24

Good

👏 Good predicted performance.

🔗 Student Performance Analyser • Academic Intelligence System • Python • Pandas • Plotly • Scikit-learn • Streamlit

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)